THE LANGUAGE OF CHEMISTRY

📖 1. Core Definitions & Key Terminology 📝

- **Chemistry:** A branch of science in which we study about matter, i.e., what is matter made up of, its nature, structure, and the changes that are observed when it is subjected to different conditions. 🧪
- **Molecule:** The smallest particle of matter which is capable of independent existence. 🧩
- **Pure Substance:** A substance which is made up of molecules containing the same kind of atoms; it has a definite composition and a definite set of properties (e.g., carbon dioxide gas). 💎
- **Element:** A simple and pure form of matter which cannot be decomposed into simpler substances. A pure substance in which a molecule is made up of atoms of the same kind. 🧱
- **Atom:** The smallest particle of an element which may or may not exist independently but can take part in a chemical reaction. ⚛️
- **Atomicity:** The number of atoms in a molecule of an element. 🔢
- **Compound:** A substance formed by the chemical combination of two or more elements in a fixed proportion (e.g., sodium chloride, water, ammonia). 🔗
- **Mixtures:** Impure substances which are formed by mixing two or more pure substances (elements or compounds) in any proportion, such that they do not undergo any chemical change and retain their individual properties. 🌪️
- **Symbol:** The short form that stands for the atom of a specific element or the abbreviation used for the name of an element. 🔤

💡 2. The "Why", Scientific Reasons & Extra Facts 🧠

- **Why are molecules considered the basic units of matter?** This is because a molecule is the smallest particle of matter that is capable of independent existence. 🔬

- **Universal Element Distribution:** At present, there are 118 elements known to us. Out of these, only two elements—hydrogen and helium—make up the majority of the known mass of the universe, with all other elements making up only 1% of the mass. 🌌
- **Why are compound properties different from their constituents?** The properties of compounds are entirely different from those of its constituent elements. For example, sodium is a highly reactive poisonous metal and chlorine is a greenish yellow poisonous gas, but the compound formed by their chemical combination (sodium chloride) is not poisonous and is actually used in our food. 🧂
- **Historical progression of symbols:**
    - Before 1600 A.D., alchemists tried to represent substances using pictographic symbols, such as a triangle [Δ] for the earth, and a crescent for silver. 🌙
    - Dalton used other types of symbols (e.g., a circle for an oxygen atom, a circle with a dot in its centre for hydrogen). ⭕
    - The universal system of symbols and formulae used today was invented by Jons Jakob Berzelius, a Swedish chemist, who suggested using the initial letter of an element's name written in capitals. SE
- **Role of IUPAC:** The International Union of Pure and Applied Chemistry (IUPAC) approves the symbols of elements, ensuring they are universally accepted. All development and research in the field of chemistry are guided by this body. 🌐
- **Why are symbols internationally advantageous?** They do not show a language dependency. For example, the English 'magnesium' changes to *magnesio* in Spanish, *magnésio* in Greek, *magnísi* in Russian, and *maguneshiumu* in Japanese. Symbols ensure the 'language' of chemistry does not change across cultural or linguistic boundaries. 🌍
- **Note on Capitalization:** Be careful about capital and small alphabets; Co means the element cobalt, however, CO means the compound carbon monoxide. ⚠️

- 🧩 *Page-Shift Stitch:* "Elements having only one atom in their molecules are monoatomic molecules like noble gases..." (Seamlessly stitched from the bottom of page 1 to the text flow regarding atomicity). 🔗

📌 3. Rules, Principles & Characteristics ⚙️

- **Characteristics of Pure Substances (Elements & Compounds):**
    - They are homogeneous, meaning their composition is uniform throughout their bulk. 🥛
    - They have a definite composition and a definite set of properties (such as boiling point, melting point, density, etc.). 🌡️
- **Characteristics of Compounds:**
    - They cannot be broken down by any physical means. 🔨🚫
- **Types of Mixtures:**
    - **Homogeneous:** Uniform composition (e.g., salt solution). 💧
    - **Heterogeneous:** Non-uniform distribution of the different types of molecules (e.g., muddy water). 🟤
- **Rules for Assigning Modern Chemical Symbols:**
    1. Usually the first letter of its name in English or Latin, written in capital (e.g., Sulphur is 'S', Hydrogen is 'H'). 🅰️
    2. If the first letter of more than one element is the same, it is represented by the first letter (capital) followed by some prominent letter (small) from their English names. Example: Carbon (C), Cobalt (Co), Calcium (Ca), Cadmium (Cd), Chromium (Cr), Chlorine (Cl). 🔡
    3. Taken from Latin/German names (e.g., Tungsten is 'W' from German *Wolfram*; Copper is 'Cu' from Latin *Cuprum*). 🏛️
- **Significance of a Symbol:** A symbol is not merely an abbreviation; it represents:
    - (i) Name of the element. 🏷️
    - (ii) One atom of the element. ⚛️

- (iii) A definite mass of the element (equal to atomic mass expressed in grams). ⚖️

📊 4. Master Data, Configurations, Formulas & Tables 🧪

- **Classification of Molecules Based on Atomicity:**
  - **Monoatomic (1 atom):** Noble gases (Helium, Neon, Argon, Krypton, Xenon, Radon), metals, and carbon. 🎈
  - **Diatomic (2 atoms):** Hydrogen (H_2), Oxygen (O_2), Nitrogen (N_2), Chlorine (Cl_2). 👫
  - **Triatomic (3 atoms):** Ozone (O_3). 🔺
  - **Tetratomic (4 atoms):** Phosphorus (P_4). ⬜
  - **Octatomic (8 atoms):** Sulphur (S_8). 🛑
- **Historical Dalton's Symbols (Textual Interpretation from Annotations):**
  - Hydrogen: Circle with a dot.
  - Carbon: Shaded/dark circle.
  - Oxygen: Empty circle.
  - Platinum: Circle with 'P'.
  - Iron: Circle with 'I'.
  - Copper: Circle with 'C'.
  - Silver: Circle with 'S'.
  - Gold: Circle with 'G'.
  - Lead: Circle with 'L'.
- **Table 1.1: Symbols of certain elements based on their Latin names**

Name of element	Symbol	Latin name
1. Gold	Au	Aurum

Name of element	Symbol	Latin name
2. Silver	Ag	Argentum
3. Mercury	Hg	Hydrargyrum
4. Copper	Cu	Cuprum
5. Lead	Pb	Plumbum
6. Iron	Fe	Ferrum
7. Sodium	Na	Natrium
8. Potassium	K	Kalium
9. Tin	Sn	Stannum
10. Antimony	Sb	Stibium
11. Tungsten	W	Wolfram *(German name)*

📖 1. Core Definitions & Key Terminology 📝

- **Valency (Traditional Definition):** A measure of an element's combining power with other elements. It is the property of an element that determines the number of other atoms with which an atom of the element can combine. 🤝
- **Valency (Standardized Definition):** Measured in terms of hydrogen or oxygen atoms. It is the number of hydrogen atoms or *double* the number of oxygen atoms that can combine with it. 📏
- **Valency of an Element or Radical:** The number of hydrogen atoms that will combine with or displace one atom of that element or radical. 🔄

- **Valency (Modern Definition):** The number of electrons that an atom can lose, gain, or share during a chemical reaction is called its valency. ⚡
- **Ion:** A positively or negatively charged atom (or group of atoms) which is formed by the loss or gain of electron(s) by an atom (or group of atoms). 🔋
- **Valence Electrons:** Electrons in the outermost shell (also known as the valence shell) of the atom. 🪐
- **Cations:** Positive ions (formed by losing electrons). ➕
- **Anions:** Negative ions (formed by gaining electrons). ➖
- **Variable Valency:** A phenomenon where certain elements exhibit more than one valency. 🎭
- **Stock Notation:** A method in chemical nomenclature that uses a Roman numeral to represent valency (e.g., Iron (II) chloride). 🏛️
- **Formula (Molecular / Chemical Formula):** The symbolic representation of a molecule. It employs symbols to denote the molecule of an element or of a compound. 🧪
- **Radical:** An atom or a group of atoms of the same or of different elements that behaves as a single unit with a positive or negative charge. ⚛️
- **Basic Radicals:** Radicals with a positive charge (cations). They tend to accept electrons to attain a stable configuration. ➕
- **Acidic Radicals:** Radicals with a negative charge (anions). They are attracted towards positive charges. ➖

💡 2. The "Why", Scientific Reasons & Extra Facts 🧠

- **Why is Hydrogen used as the standard for valency?** Since no other element has a combining capacity less than that of hydrogen, its valency is taken to be one (1) and is considered a standard. 📏
- **Why is an atom normally electrically neutral?** Every atom normally contains an equal number of electrons and protons. The 1 unit negative

charge of each electron balances the 1 unit positive charge of each proton (neutrons have no charge). ⚖️

- **How does an atom become an ion?** If one or more electrons are either removed from or added to an atom, the atom has either less or more electrons than normal, respectively. It gets an overall electric charge, becoming an ion. ⚡
- **Why do atoms lose, gain, or share electrons?** To attain stability. Atoms have to have eight electrons (attain an octet) in their outermost shell (except hydrogen, which attains stability with 2 electrons). 🎯
- **Why do some elements exhibit variable valency?** An atom of an element can sometimes lose more electrons than are present in its valence shell; i.e., there is a loss of electrons from the penultimate (next to outermost) shell too. Therefore, such an element exhibits variable valency. 🎭
- **Why are basic radicals positively charged?** They are formed when a base loses its hydroxide ion (OH^-). For example, Na^+ is formed by the removal of OH^- from base sodium hydroxide ($NaOH$). Being positively charged, these cations are attracted towards negative charges. 🧲
- **Why are acidic radicals negatively charged?** They are formed by the removal of hydrogen ion (H^+) from acids. For example, Cl^- is formed by removing H^+ from hydrochloric acid (HCl). These anions are attracted towards positive charges. 🧲
- **Significance of a molecular formula:** It quantitatively represents: (i) both the molecule and the molecular mass of the compound, (ii) the respective numbers of different atoms present in one molecule, and (iii) the ratios of the respective masses of the elements present in the compound. 📊
- **Example Breakdown of CO_2 Formula:**
    - Molecular formula is CO_2.
    - Each molecule contains 1 carbon atom joined by chemical bonds with 2 oxygen atoms.
    - Molecular mass is 44 (atomic mass of $C = 12$, atomic mass of $O = 16$).

    - Carbon and oxygen combine in the mass ratio of $12:32$ or $3:8$.
- ✍️ *Handwritten Margin Note:* "→ least combining capacity" (Written pointing to the explanation of why Hydrogen is the standard for valency). 📝
- ✍️ *Handwritten Margin Note:* "X_O" (Circled, written next to the tetravalent element Lead (Pb)). 📝
- ✍️ *Handwritten Margin Note:* "⋆" (A star inside a circle, drawn next to the text: "Thus by looking at a formula, we understand the ratio in which the different atoms are united..."). 📝
- ✍️ *Handwritten Margin Note:* "Fe_3O_4 → triferric tetraoxide" (Written at the bottom of the page). 📝
- 🧩 *Page-Shift Stitch:* "Non-metals like nitrogen, phosphorus and sulphur also show variable valency. Nitrogen and phosphorus exhibit valencies of 3 and 5 while sulphur exhibits valencies of 2,4 and 6." (Seamlessly stitched from the bottom of page 5 to the top of page 6). 📝

📌 3. Rules, Principles & Characteristics ⚙️

- **Valency Calculation (Hydrogen Standard):**
    - In HCl, 1 atom of Cl combines with 1 atom of H. Valency of $Cl = 1$.
    - In NH_3, 1 atom of N combines with 3 atoms of H. Valency of $N = 3$.
    - In CH_4, 1 atom of C combines with 4 atoms of H. Valency of $C = 4$.
- **Valency Calculation (Oxygen Standard):**
    - In Na_2O, 1 atom of O combines with 2 atoms of Na. Valency of $Na = 2 \times (1/2) = 1$.
    - In CO_2, 1 atom of C combines with 2 atoms of O. Valency of $C = 2 \times 2 = 4$.
- **Rules for Ion Formation (Metals vs. Non-metals):**
    - Atoms with 1,2, or 3 valence electrons lose them to attain stability, forming monovalent, divalent, and trivalent positive ions respectively. (These are usually metals). 🥇

- Atoms with 7,6, and 5 valence electrons gain 1,2, and 3 electrons respectively to complete their octet, forming monovalent, divalent, and trivalent negative ions. (These are normally non-metals). 🪵
- *Exceptions for 4 valence electrons:* Carbon is a non-metal, Silicon and Germanium are metalloids, and the rest (like Tin, Lead) are metals.

- **Naming Variable Valencies (Old vs. New):**
  - *Old System:* Suffix "-ous" is used for the lower valency; "-ic" is used for the higher valency (e.g., Stannous, Stannic). 🏷️
  - *Modern System:* Uses Roman numerals in brackets (Stock notation) in place of trivial names (e.g., Tin (II) chloride, Tin (IV) chloride). 🏛️

- **Rules for Writing Molecular Formulas:**
  - Molecules of an element can have multiple atoms written as subscripts (e.g., $H_2, O_2, N_2, Cl_2, Br_2, I_2$).
  - In a compound, symbols of elements are placed side-by-side with subscripts denoting their fixed ratio (e.g., $NH_4Cl = 1N, 4H, 1Cl$; $Na_2CO_3 = 2Na, 1C, 3O$).
  - Numbers placed *before* the formula denote the number of molecules (e.g., $2H_2O = 2$ molecules of water). 🔢

- **Classification of Elements by Valency:**
  - *Monovalent:* Valency 1 (e.g., H, Li, Na, K, Cl, F).
  - *Divalent:* Valency 2 (e.g., Mg, Ca, Fe^{2+}, O).
  - *Trivalent:* Valency 3 (e.g., B, Al, N, P).
  - *Tetravalent:* Valency 4 (e.g., C, Si, Ge, Sn, Pb).

📊 4. Master Data, Configurations, Formulas & Tables 🧪

- **Formation of Cations (Positive Ions):**
  - $Na - e^- \rightarrow Na^+$ (monovalent) | 2,8,1 → 2,8
  - $Mg - 2e^- \rightarrow Mg^{2+}$ (divalent) | 2,8,2 → 2,8

- $Al - 3e^- \rightarrow Al^{3+}$ (trivalent) | $2,8,3 \rightarrow 2,8$

- **Formation of Anions (Negative Ions):**
  - $Cl + e^- \rightarrow Cl^-$ (monovalent) | $2,8,7 \rightarrow 2,8,8$
  - $O + 2e^- \rightarrow O^{2-}$ (divalent) | $2,6 \rightarrow 2,8$
  - $N + 3e^- \rightarrow N^{3-}$ (trivalent) | $2,5 \rightarrow 2,8$

- **Relationship Between Valence Electrons and Valency:**

No. of electrons in outermost shell	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Valency	1	2	3	4	3	2	1	0

- **Table 1.3: Examples of variable valency**

Element	**Valency**	**Name of compound formed**	**Formula**
Iron	2	Ferrous (Iron (II) oxide)	FeO
	3	Ferric (Iron (III) oxide)	Fe_2O_3
Copper	1	Cuprous (Copper (I) oxide)	Cu_2O
	2	Cupric (Copper (II) oxide)	CuO
Mercury	1	Mercurous (Mercury (I) oxide)	Hg_2O
	2	Mercuric (Mercury (II) oxide)	HgO
Lead	2	Plumbous (Lead (II) oxide)	PbO
	4	Plumbic (Lead (IV) oxide)	PbO_2

Element	Valency	Name of compound formed	Formula
Tin	2	Stannous (Tin (II) chloride)	$SnCl_2$
	4	Stannic (Tin (IV) chloride)	$SnCl_4$
Nitrogen	3	Ammonia	NH_3
	5	Dinitrogen pentoxide	N_2O_5
Phosphorus	3	Phosphorus trichloride	PCl_3
	5	Phosphorus pentachloride	PCl_5
Sulphur	2	Hydrogen sulphide	H_2S
	4	Sulphur dioxide	SO_2
	6	Sulphur trioxide	SO_3

📖 1. Core Definitions & Key Terminology 📝

- **Simple Radical:** A radical is called a simple radical when it is an atom only, like sodium (Na^+) and magnesium (Mg^{2+}).
- **Compound or Complex Radical:** A radical is known as a compound or complex radical when it is made up of a group of two or more different atoms.
- **Polyatomic Radical:** Another term for a compound or complex radical. For example, sulphate (SO_4^{2-}) is a polyatomic radical made up of one sulphur atom and four oxygen atoms.
- **Neutralisation:** The process where an acid reacts with a base to produce salt and water.

💡 2. The "Why", Scientific Reasons & Extra Facts 🧠

- **Why are acidic and basic radicals highly reactive?** Acidic (anions) and basic (cations) radicals are highly reactive because they are unstable, charged intermediates formed from the dissociation of salts, acting as electron donors or acceptors to achieve stable configuration.
- **How do salts split in water?** When a salt is dissolved in water, it splits up into its constituent radicals.
- **Where do radicals in a salt come from?** In a salt (like potassium chloride), the electropositive (or basic) radical (e.g., K^+) has been contributed by the base (potassium hydroxide), while the electronegative (or acidic) radical (e.g., Cl^-) has been contributed by the acid (hydrochloric acid).
- **What determines a radical's chemical formulae formation?** The charge on a radical is usually its valency. Radicals have their own combining power (valency), and it is according to this combining power that they form chemical formulae.
- **Nomenclature History Note:** The suffixes *-ous* and *-ic* were used in old nomenclature systems to represent lower and higher valencies respectively.
- ✍️ *Handwritten Margin Note:* "Au" (Written above the 'Trivalent electropositive' column heading).
- ✍️ *Handwritten Margin Note:* "$Fe(CN)_6^{-3}$ → ferricyanide three" (Written next to the Divalent electronegative table, noting the trivalent radical).
- ✍️ *Handwritten Margin Note:* "four" (Written with an arrow pointing to the Ferrocyanide radical in the Tetravalent electronegative section).

📌 3. Rules, Principles & Characteristics ⚙️

- **Rule for Naming Compounds (General Steps):**
    1. Identify the basic and acidic radicals in the formula.
    2. Write the name of the compound in the same order of radicals as given in the formula.
- **Rule for Compounds with a Metal and a Non-metal:**

- The metal is written first followed by the non-metal, and the suffix *ide* is added to the non-metal.
- *Example:* Calcium + Nitrogen → Calcium nitride (Ca_3N_2).

- **Rule for Compounds of Variable Valencies (Metals):**
  - For elements like gold, iron, copper, mercury, lead, and tin that exhibit more than one valency, the latest IUPAC system dictates that the valency is written in Roman numbers in brackets with the basic radical.
  - *Example:* Gold (I) sulphate is Au_2SO_4 (Aurous sulphate). Gold (III) chloride is $AuCl_3$ (Auric chloride).
- **Rule for Compounds with Non-metals having Variable Valency:**
  - The prefix *tri* or *tetra*, *penta* or etc. is added, as the case may be, to the second non-metal.
  - *Example:* PCl_3 is phosphorus trichloride and PCl_5 is phosphorus pentachloride.
- **Rule for Compounds with Different Numbers of Oxygen Atoms:**
  - Oxygen is represented at the end of the formula.
  - The name of the compound depends on the number of oxygen atoms present in the compound.
  - The prefix 'hypo' is used if the number of oxygen atoms is less than 2 (e.g., $KClO$ - Potassium hypochlorite).

📊 4. Master Data, Configurations, Formulas & Tables 🧪

- **Table 1.4: List of some common electrovalent positive ions (basic radicals)**

Monovalent electropositive	**Formula**	**Divalent electropositive**	**Formula**	**Trivalent electropositive**	**Formula**
1. Ammonium	NH_4^+	1. Argentic [Silver(II)]	Ag^{2+}	1. Aluminium	Al^{3+}
2. Aurous [Gold (I)]	Au^+	2. Barium	Ba^{2+}	2. Arsenic	As^{3+}
3. Argentous [Silver (I)]	Ag^+	3. Calcium	Ca^{2+}	3. Auric [Gold (III)]	Au^{3+}
4. Cuprous [Copper (I)]	Cu^+	4. Cupric [Copper (II)]	Cu^{2+}	4. Bismuth	Bi^{3+}
5. Hydrogen	H^+	5. Ferrous [Iron (II)]	Fe^{2+}	5. Chromium	Cr^{3+}
6. Lithium	Li^+	6. Magnesium	Mg^{2+}	6. Ferric [Iron (III)]	Fe^{3+}
7. Sodium	Na^+	7. Manganese	Mn^{2+}		
8. Potassium	K^+	8. Mercuric [Mercury (II)]	Hg^{2+}	**Tetravalent electropositive**	
9. Mercurous	Hg^+	9. Nickel	Ni^{2+}	1. Plumbic [Lead (IV)]	Pb^{4+}

Monovalent electropositive	Formula	Divalent electropositive	Formula	Trivalent electropositive	Formula
[Mercury (I)]					
		10. Plumbous [Lead (II)]	Pb^{2+}	2. Platinic [Platinum (IV)]	Pt^{4+}
		11. Platinous [Platinum (II)]	Pt^{2+}	3. Stannic [Tin (IV)]	Sn^{4+}
		12. Stannous [Tin (II)]	Sn^{2+}		
		13. Zinc	Zn^{2+}		

- **Table 1.5: List of some common electrovalent negative ions (acid radicals)**

Monovalent electronegative	Formula	Divalent electronegative	Formula	Trivalent electronegative	Formula
1. Acetate	CH_3COO^-	1. Carbonate	CO_3^{2-}	1. Arsenate	AsO_4^{3-}
2. Bicarbonate or	HCO_3^-	2. Dichromate	$Cr_2O_7^{2-}$	2. Nitride	N^{3-}

Monovalent electronegative	**Formula**	**Divalent electronegative**	**Formula**	**Trivalent electronegative**	**Formula**
Hydrogen carbonate					
3. Bisulphide or Hydrogen sulphide	HS^-	3. Oxide	O^{2-}	3. Aluminate	AlO_3^{3-}
4. Bisulphate or Hydrogen sulphate	HSO_4^-	4. Peroxide	O_2^{2-}	4. Arsenite	AsO_3^{3-}
5. Bisulphite or Hydrogen sulphite	HSO_3^-	5. Sulphate	SO_4^{2-}	5. Phosphide	P^{3-}
6. Bromide	Br^-	6. Sulphite	SO_3^{2-}	6. Phosphite	PO_3^{3-}
7. Chloride	Cl^-	7. Sulphide	S^{2-}	7. Phosphate	PO_4^{3-}
8. Permanganate	MnO_4^-	8. Silicate	SiO_3^{2-}	8. Borate	BO_3^{3-}

Monovalent electronegative	**Formula**	**Divalent electronegative**	**Formula**	**Trivalent electronegative**	**Formula**
9. Fluoride	F^-	9. Thiosulphate	$S_2O_3^{2-}$		
10. Hydride	H^-	10. Zincate	ZnO_2^{2-}	**Tetravalent electronegative**	
11. Hydroxide	OH^-	11. Plumbite	PbO_2^{2-}	1. Carbide	C^{4-}
12. Iodide	I^-	12. Stannate	SnO_3^{2-}	2. Ferrocyanide	$Fe(CN)_6^{4-}$
13. Cyanide	CN^-	13. Manganate	MnO_4^{2-}		
14. Nitrate	NO_3^-	14. Chromate	CrO_4^{2-}		
15. Nitrite	NO_2^-	15. Oxalate	$(COO)_2^{2-}$		
16. Chlorite	ClO_2^-				
17. Hypochlorite	ClO^-				

Monovalent electronegative	Formula	Divalent electronegative	Formula	Trivalent electronegative	Formula
18. Chlorate	ClO_3^-				
19. Perchlorate	ClO_4^-				
20. Meta-aluminate	AlO_2^-				

- **Acidic Radicals showing Variable Valency Table**

Acidic radical	Compound	Name of the compound
O^{2-}	Na_2O	Sodium oxide
O_2^{2-}	Na_2O_2	Sodium peroxide
MnO_4^-	$KMnO_4$	Potassium permanganate
MnO_4^{2-}	K_2MnO_4	Potassium manganate
$[Fe(CN)_6]^{4-}$	$K_4[Fe(CN)_6]$	Potassium ferrocyanide
$[Fe(CN)_6]^{3-}$	$K_3[Fe(CN)_6]$	Potassium ferricyanide

📖 1. Core Definitions & Key Terminology 📝

- **Free Radicals:** Atoms or molecules or ions that contain at least one unpaired valence electron.

- **Homolytic Fission (or Homolysis):** A process where a covalent bond breaks evenly, resulting in each atom receiving one unpaired electron and becoming a free radical.
- **Oxidative Stress:** A harmful condition that occurs when free radicals are produced in excess, leading to the damage of healthy cells.
- **Chemical Equation:** The symbolic representation of a chemical reaction using the symbols and formulae of the substances involved in the reaction.
- **Reactants:** The substances that are involved in a given chemical reaction.
- **Products:** The substances formed as a result of the chemical reaction.
- **Skeleton Equation:** An equation that represents a chemical change but is unbalanced; the total number of atoms of each element on the two sides are not equal.
- **Balanced Equation:** An equation in which the total number of atoms of each element in the reactants on the left side of the equation is exactly the same as the number of atoms in the products formed on the right side.
- **Law of Conservation of Matter:** A scientific law stating that matter is neither created nor destroyed in the course of a chemical reaction.

💡 2. The "Why", Scientific Reasons & Extra Facts 🧠

- **Why are they called 'free' radicals?** They are called 'free' because they can exist independently, and their unpaired electron is free to react easily with other atoms or molecules.
- **Why are free radicals exceptionally reactive?** Normally, electrons exist in pairs within a molecule; when a particle has an electron left alone without a partner, it becomes highly unstable and exceptionally reactive, making free radicals short-lived species.
- **Why is the Hydroxyl radical ($OH\cdot$) called the "detergent of the atmosphere"?** It plays a vital role in removing air pollutants, including methane (CH_4), carbon monoxide, and other harmful gases. ✍️ *Handwritten Margin Note:* "imp" (circled next to this definition).

- **How does the Nitrate radical ($NO_3 \cdot$) function?** It is especially active at night, reacting with harmful gases and pollutants in the air and changing them into forms that can be washed out by rain.
- **Why is the Chlorine radical ($Cl \cdot$) dangerous?** It is released from chlorofluorocarbons (CFCs) and related compounds in sunlight, taking part in atmospheric reactions that destroy the ozone layer.
- **How are free radicals produced in the human body?** They are produced naturally during normal biological processes such as breathing, digestion, and cellular metabolism (when oxygen is used to release energy).
- **What are the beneficial effects of free radicals?** In small, controlled amounts, they help the immune system destroy bacteria and viruses, play a role in cell signalling, and assist in healing and defense mechanisms.
- **What are the harmful effects of free radicals?** Excess radicals cause oxidative stress which can damage cell membranes, harm proteins and enzymes, affect DNA and genetic material, speed up aging processes, and increase the risk of various diseases.
- **How can free radicals be reduced or eliminated?** Restricting the use of CFCs reduces atmospheric chlorine radicals. In the body, substances called antioxidants (like vitamin C and vitamin E) help neutralize excess free radicals.
- **Why must a chemical equation be balanced?** An equation must be balanced in order to comply with the "Law of Conservation of Matter". An unbalanced equation would incorrectly imply that atoms have been created or destroyed.
- **Why represent reactants and products in molecular forms?** They are written in molecular forms because their atomic forms are usually neither stable nor capable of separate existence.
- **Note on Trivial Names:** Certain compounds do not follow any systematic naming rule; such widely accepted names are called trivial or common names (e.g., Nitrogen trihydride is ammonia NH_3, Hydrogen monoxide is water H_2O).

- ✍️ *Handwritten Margin Note:* A bracketed box containing "$[Hg^{+1}, Hg^{+2}]$" is written at the bottom of the page near the exercises.

📌 3. Rules, Principles & Characteristics ⚙️

- **Rules for Naming Acids (Binary vs. Polyatomic):**
    - *Binary Acids:* Names of acids composed of two elements are given by adding the prefix 'hydro' and the suffix 'ic' to the name of the second element (e.g., HCl is hydrochloric acid, HF is hydrofluoric acid).
    - *Polyatomic Acids:* Names of acids containing radicals of polyatomic groups are based on the second element present; the prefix 'hydro' is NOT used (e.g., H_2SO_4 is sulphuric acid, HNO_3 is nitric acid).
    - *Oxygen Quantity Suffixes (Acids):* If the number of oxygen atoms is less, the suffix 'ous' is used instead of 'ic' (e.g., H_2SO_3 is sulphurous acid, HNO_2 is nitrous acid).
- **Rules for Oxygen Quantity Suffixes/Prefixes (Salts):**
    - Suffix 'ite': Used if the number of oxygen atoms is 2 (e.g., $KClO_2$ - Potassium chlorite).
    - Suffix 'ate': Used if the number of oxygen atoms is 3 (e.g., $KClO_3$ - Potassium chlorate).
    - Prefix 'per': Used when the number of oxygen atoms is more than 3 (e.g., $KClO_4$ - Potassium perchlorate).
- **Procedure to Find Valency from a Formula:**
    1. Write the given formula (e.g., NO_2).
    2. Interchange the subscript and write it as a superscript (e.g., N^2O^1).
    3. Multiply both superscripts by the known valency of oxygen (which is 2) or hydrogen (which is 1) (e.g., $N^{2\times2}O^{1\times2}$).
    4. The result gives the valency of the elements (e.g., N^4O^2; thus valency of nitrogen is 4).

- **Steps Involved in Writing a Chemical Equation:**
    1. Write the symbols or formulae of the reactants on the left-hand side, with a (+) sign between them.
    2. Write the symbols or formulae of the products on the right-hand side, with a (+) sign between them.
    3. Put the sign of an arrow (→) or (⇌) in between the reactant side and the product side to show the direction of the reaction.
    4. Represent the reactants and products in their molecular forms.
- **Steps for Balancing (Hit and Trial Method):**
    1. Count the number of times (frequency) an element occurs on either side.
    2. The element with the least frequency of occurrence is balanced first.
    3. When two or more elements have the same frequency, the metallic element is balanced first.

📊 4. Master Data, Configurations, Formulas & Tables 🧪

- **Nomenclature of Acids and Salts Table:**

(Note: Handwritten scribbles circle $HClO$ and $HClO_4$ in the printed table).

Acid	**Acidic Radical**	**Compound formed**
$HClO$ (Hypochlorous acid)	ClO^- (Hypochlorite)	$NaClO$ (Sodium hypochlorite)
$HClO_2$ (Chlorous acid)	ClO_2^- (Chlorite)	$NaClO_2$ (Sodium chlorite)
$HClO_3$ (Chloric acid)	ClO_3^- (Chlorate)	$NaClO_3$ (Sodium chlorate)
$HClO_4$ (Perchloric acid)	ClO_4^- (Perchlorate)	$NaClO_4$ (Sodium perchlorate)

- **Free Radical Fission Example:**
  - $Cl_2 \xrightarrow{\text{sunlight}} Cl\cdot + Cl\cdot$
- **Types of Chemical Reactions based on Reactant/Product Quantities:**
  - *(i) One reactant → two or more products:* $CaCO_3 \rightarrow CaO + CO_2$ OR $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$
  - *(ii) Two reactants → one product:* $NH_3 + HCl \rightarrow NH_4Cl$ OR $N_2 + 3H_2 \rightarrow 2NH_3$ OR $2Cu + O_2 \rightarrow 2CuO$
  - *(iii) Two reactants → two products:* $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$ OR $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$
  - *(iv) Two reactants → three or more products:*
    - $Cu + 2H_2SO_4 \rightarrow CuSO_4 + 2H_2O + SO_2$
    - $2KMnO_4 + 16HCl \rightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2$ *(Handwritten Note: The entire $KMnO_4$ equation is circled).*
- **Step-by-Step Balancing Examples (Hit and Trial):**
  - **Example 1 Skeletal:** $Cu + H_2SO_4 \rightarrow CuSO_4 + SO_2 + H_2O$
    - *Step 1 (Count):* Reactants (Cu: 1, H: 2, S: 1, O: 4), Products (Cu: 1, H: 2, S: 2, O: 7).
    - *Step 2 (Balance S):* Multiply H_2SO_4 by 2. → $Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + H_2O$
    - *Step 3 (Balance H):* Multiply H_2O by 2. → $Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$ (Balanced).
  - **Example 2 Skeletal:** $Fe + H_2O \rightarrow Fe_2O_3 + H_2$
    - *Step 1 (Count):* Reactants (Fe: 1, H: 2, O: 1), Products (Fe: 2, H: 2, O: 3).
    - *Step 2 (Balance Fe):* Multiply reactant Fe by 2. → $2Fe + H_2O \rightarrow Fe_2O_3 + H_2$.

📖 1. Core Definitions & Key Terminology 📝

- **Partial Equation Method:** A method of balancing complex chemical equations by supposing the complex reaction to have taken place in steps,

writing equations for these individual steps, and then adding them together.

- **Information Conveyed by a Balanced Chemical Equation:** A balanced equation tells us about the actual result of the chemical change, the reactants involved, the products formed, the number of molecules of each substance taking part, the chemical composition of the respective molecules, and the molecular mass relationships.
- **Limitations of a Chemical Equation:** A standard chemical equation does not tell us the physical state of the reactants/products, the time taken for completion, whether heat is given out or absorbed, the respective concentrations, the rate of reaction, whether the reaction is reversible/irreversible, or whether the reaction is actually completed.

💡 2. The "Why", Scientific Reasons & Extra Facts 🧠

- **Why does the Hit and Trial method have limitations?** It takes time to balance complicated equations, and the mechanism (the steps of the reaction, i.e., how the reaction has taken place) of the reaction is not clear.
- **Why is the Partial Equation method used?** Balancing can be done more easily by supposing a complex reaction to have taken place in steps.
- **How does an equation prove the law of conservation of mass?** It shows that the total mass of the substances on either side of the equation is the same. For example, in $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$, 178 g of reactants produce exactly 178 g of products.
- **Standard Temperature and Pressure (STP) Fact:** On the basis of experiments, it is found that 22.4 litres of every gas at 760 mm pressure and a temperature of $0°C$ (STP) weighs the same as its molar mass (molecular mass expressed in grams).
- **Why are symbols added to equations?** A chemical equation can be made more informative by writing additional information to it, overcoming some of its inherent limitations.
- ✍️ *Handwritten Margin Note:* A star inside a circle is drawn next to the heading "Information conveyed by a balanced chemical equation".

- ✍️ *Handwritten Margin Note:* The words "Not in syllabus" are written faintly under the "Balancing by partial equation method" heading.
- 🧩 *Page-Shift Stitch:* "Step 4: Now there are 6 hydrogen atoms on the reactant side, and only 2 hydrogen atoms on the product side. To balance the hydrogen atoms, write 3 before hydrogen on the product side.".

📌 3. Rules, Principles & Characteristics ⚙️

- **Rules for Making a Chemical Equation More Informative:**
  - **(a) Physical States:** Can be provided by using the letters (s) for solid, (l) for liquid, (g) or upward arrow (↑) for gas, and (aq) for solution in water.
  - **(b) Concentration:** Concentration of acids can also be added (e.g., $dil.$ for dilute, $conc.$ for concentrated).
  - **(c) Heating:** Heating during the reaction can be shown by a sign of delta (Δ) over the arrow.
  - **(d) Precipitates:** Any reaction in which a precipitate is formed can be represented by a downward arrow (↓).
  - **(e) Reversibility:** Reactions which are reversible can be shown by the sign of reversibility (⇌).
  - **(f) Heat Evolution/Absorption:** Information is provided by adding the term 'heat' (e.g., +Heat or −Heat).
  - **(g) Reaction Conditions:** Information regarding temperature, pressure, catalyst, etc., is provided above and or below the arrow (→) separating reactants and products.

📊 4. Master Data, Configurations, Formulas & Tables 🧪

- **Step-by-Step Balancing Examples (Hit and Trial - Complex):**
  - **Example 4 Skeletal:** $As_2O_3 + SnCl_2 + HCl \rightarrow SnCl_4 + As + H_2O$
    - *Step 2 (Balance As and O):* Multiply As by 2 and H_2O by 3 → $As_2O_3 + SnCl_2 + HCl \rightarrow SnCl_4 + 2As + 3H_2O$
    - *Step 3 (Balance H):* Multiply HCl by 6 → $As_2O_3 + SnCl_2 + 6HCl \rightarrow SnCl_4 + 2As + 3H_2O$

        - *Step 4 (Balance Cl):* Multiply $SnCl_2$ and $SnCl_4$ by 3 → $As_2O_3 + 3SnCl_2 + 6HCl \rightarrow 3SnCl_4 + 2As + 3H_2O$ (Balanced)
    - **Example 5 Skeletal:** $K_2Cr_2O_7 + HCl \rightarrow KCl + CrCl_3 + H_2O + Cl_2$
        - *Step 2 Sequence:* Equalize oxygen ($7H_2O$), equalize hydrogen ($14HCl$), equalize K and Cr ($2KCl$ and $2CrCl_3$), equalize Cl ($3Cl_2$).
        - *Final Balanced:* $K_2Cr_2O_7 + 14HCl \rightarrow 2KCl + 2CrCl_3 + 7H_2O + 3Cl_2$
- **Step-by-Step Balancing Example (Partial Equation Method):**
    - *Reaction:* Liberation of iodine from potassium iodide by reacting it with hydrogen peroxide.
    - *Step 1:* H_2O_2 decomposes to give water + nascent oxygen. → $H_2O_2 \rightarrow H_2O + [O]$
    - *Step 2:* Nascent oxygen oxidizes potassium iodide in the presence of water to give iodine and potassium hydroxide. → $2KI + H_2O + [O] \rightarrow 2KOH + I_2$
    - *Result:* Adding equations and cancelling intermediates (H_2O and $[O]$). → $2KI + H_2O_2 \rightarrow 2KOH + I_2$
- **Informative Equation Examples:**
    - *Physical states:* $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$
    - *Concentration:* $Zn(s) + H_2SO_4(dil) \rightarrow ZnSO_4(aq) + H_2(g)$
    - *Concentration (different product):* $Zn(s) + 2H_2SO_4(conc.) \rightarrow ZnSO_4(aq) + 2H_2O(l) + SO_2(g)$
    - *Heating:* $2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$
    - *Precipitate:* $AgNO_3(aq) + HCl(aq) \rightarrow HNO_3(aq) + AgCl\downarrow$
    - *Reversibility:* $3Fe + 4H_2O \rightleftharpoons Fe_3O_4 + 4H_2$
    - *Conditions (Catalyst/Temp/Pressure):* $N_2 + 3H_2 \xrightarrow[200-900\ atm]{Fe+Mo,\ 450^\circ C} 2NH_3 +$ heat

📖 1. Core Definitions & Key Terminology 📝

- **Atomic Mass Unit (a.m.u. or u):** Defined as $1/12$ the mass of a carbon atom $C-12$. (The mass of an atom of carbon-12 isotope was given the atomic mass of 12 units).
- **Relative Atomic Mass (RAM) or Atomic Weight:** The number of times one atom of the element is heavier than $1/12$ times of the mass of an atom of carbon-12.
- **Relative Molecular Mass (RMM) or Molecular Weight:** The number that represents how many times one molecule of the substance is heavier than $1/12$ of the mass of an atom of carbon-12.
- **Percentage Composition:** The percentage by weight of each element present in a compound.
- **Empirical Formula:** The formula which gives the simplest whole number ratio of atoms of different elements present in one molecule of the compound.
- **Empirical Formula Mass:** The sum of atomic masses of various elements present in the empirical formula.

💡 2. The "Why", Scientific Reasons & Extra Facts 🧠

- **Why is indirect measurement used for atomic mass?** Atoms being extremely small, cannot be seen or weighed directly. Indirect methods of physics have enabled us to know the absolute mass of nearly all kinds of atoms.
- **Why was Carbon-12 chosen as the standard?** In the beginning, the mass of the hydrogen atom was chosen as a unit. In 1961, carbon-12 was finally selected because its adoption least affected the values of the atomic mass of the various elements on the old standard.
- **What is the distinction between actual mass and atomic mass?** Atomic mass of an element is not its actual mass but its relative mass. The element carbon is used as a reference for expressing the atomic masses of elements.
- **Note on Percentage Composition:** Total percentage of all the elements in a compound is 100.

- **Note on Empirical Formula Uniqueness:** Empirical formula can be same for two or more compounds. For example: Glucose ($C_6H_{12}O_6$) and acetic acid (CH_3COOH or $C_2H_4O_2$) both have the same empirical formula CH_2O.
- ✍️ *Handwritten Margin Note:* "1 amu $= 1.66 \times 10^{-24}$ kg" (Written boldly at the bottom of the page, though the printed text says grams, the handwritten note distinctly says 'kg' ending).
- ✍️ *Handwritten Margin Note:* "Mohr's salt" is underlined and circled in Example 10.
- ✍️ *Handwritten Margin Note:* A star symbol is drawn next to the definition of Empirical Formula.
- 🧩 *Page-Shift Stitch:* "The atomic mass of an element is the average mass of the atoms in it." (Stitched from the top right column of page 19).

📌 3. Rules, Principles & Characteristics ⚙️

- **Formula for Relative Atomic Mass:**
  - Relative atomic mass $= \frac{\text{Mass of 1 atom of the element}}{\frac{1}{12} \text{ the mass of one C-12 atom}}$
- **Calculation of 1 a.m.u. in grams:**
  - Experimentally it is found that one atom of C^{12} has a mass of 1.9926×10^{-23} grams.
  - On dividing this mass by 12, atomic mass unit (amu or u) is obtained.
  - 1 amu or 1 u $= 1.6605 \times 10^{-24}$ g
- **Procedure for finding Relative Molecular Mass (RMM):**
  - Obtained by adding together the relative atomic masses (atomic weights) of all the various atoms present in a molecule.
- **Formula for Percentage Composition:**
  - Percentage of an element in a compound $= \frac{\text{Total wt. of the element in one molecule}}{\text{Gram molecular weight of the compound}} \times 100$

📊 4. Master Data, Configurations, Formulas & Tables 🧪

- **Standard Atomic Masses (from examples):**

- Hydrogen (H) = 1 amu
- Carbon (C) = 12 amu
- Oxygen (O) = 16 amu
- Nitrogen (N) = 14 amu
- Sulphur (S) = 32 amu
- Copper (Cu) = 63.5 amu
- Sodium (Na) = 23 amu
- Iron (Fe) = 56 amu

- **Relative Molecular Mass Calculations (Examples):**
  - *Sulphuric Acid (H_2SO_4):* $(1 \times 2) + 32 + (16 \times 4) = 98$ amu
  - *Cane sugar ($C_{12}H_{22}O_{11}$):* $(12 \times 12) + (1 \times 22) + (16 \times 11) = 342$ amu
  - *Copper sulphate crystals ($CuSO_4 \cdot 5H_2O$):* $63.5 + 32 + (16 \times 4) + 5(2 + 16) = 249.5$ amu
  - *Ammonium sulphate ($(NH_4)_2SO_4$):* $14 \times 2 + 1 \times 8 + 32 + 16 \times 4 = 132$ amu
- **Complex Percentage Composition (Mohr's Salt Example 10):**
  - *Formula:* $(NH_4)_2SO_4 \cdot FeSO_4 \cdot 6H_2O$
  - *Total R.M.M:* 392 amu
  - *Mass percent of N:* $\frac{28}{392} \times 100 = 7.14\%$
  - *Mass percent of H:* $\frac{20}{392} \times 100 = 5.10\%$
  - *Mass percent of S:* $\frac{64}{392} \times 100 = 16.33\%$
  - *Mass percent of Fe:* $\frac{56}{392} \times 100 = 14.28\%$
  - *Mass percent of O:* $\frac{224}{392} \times 100 = 57.14\%$
  - *Percent of water of crystallization (6 molecules):* $\frac{6 \times 18}{392} \times 100 = 27.55\%$
- **Empirical Formula Examples:**

- *Hydrogen peroxide:* Actual formula = H_2O_2 → Empirical formula = HO (Ratio 1:1)
- *Glucose:* Actual formula = $C_6H_{12}O_6$ → Empirical formula = CH_2O (Ratio 1:2:1)